D04 | XOR decision block

All-case aligned observations: 31,509 | Train: 22,056 | Held-out test: 9,453

Test accuracy 0.777 | Majority baseline 0.707 | Balanced accuracy 0.825 | Macro-F1 0.765

B1 -> A_Cancelled | aligned support 9,532
B2 -> parallelGateway | aligned support 21,977

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 297 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 150.5 -> B1 [train n=3,099]

e.CreditScore <= 297 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 150.5 -> B1 [train n=7,622]

e.CreditScore <= 297 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 1 -> B2 [train n=1,496]

e.CreditScore <= 297 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 1 -> B2 [train n=545]

e.CreditScore > 297 -> B2 [train n=9,294]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.